

dsbda-lab-ass-8

April 3, 2026

1 Assignment 8

1.1 Import Required Library

```
[1]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

1.2 Read Titanic dataset into Dataframe

```
[2]: titanic = sns.load_dataset('titanic')
```

```
[3]: titanic.shape
```

```
[3]: (891, 15)
```

```
[4]: titanic.head()
```

```
[4]:   survived  pclass    sex  age  sibsp  parch   fare embarked  class \
0         0        3   male  22.0     1     0   7.2500         S   Third
1         1        1  female  38.0     1     0  71.2833         C   First
2         1        3  female  26.0     0     0   7.9250         S   Third
3         1        1  female  35.0     1     0  53.1000         S   First
4         0        3   male  35.0     0     0   8.0500         S   Third
```

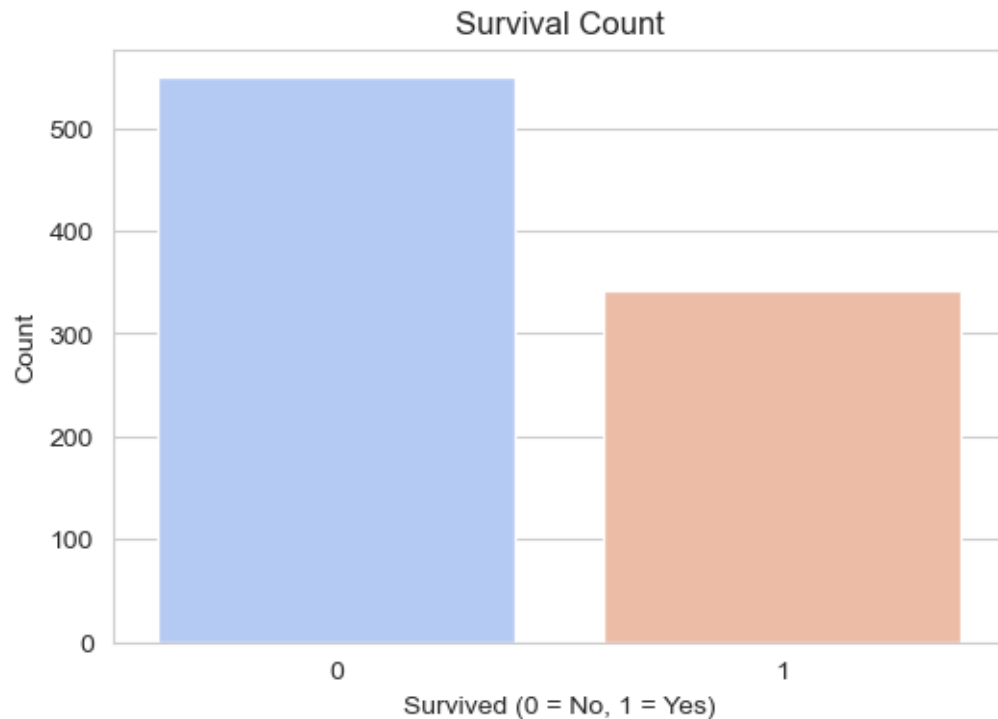
```
   who  adult_male  deck  embark_town  alive  alone
0  man         True  NaN  Southampton    no  False
1 woman        False   C   Cherbourg   yes  False
2 woman        False  NaN  Southampton   yes   True
3 woman        False   C   Southampton   yes  False
4  man         True  NaN  Southampton    no   True
```

1.3 Set the style for the plots

```
[5]: sns.set_style("whitegrid")
```

1.4 1. Survival countplot

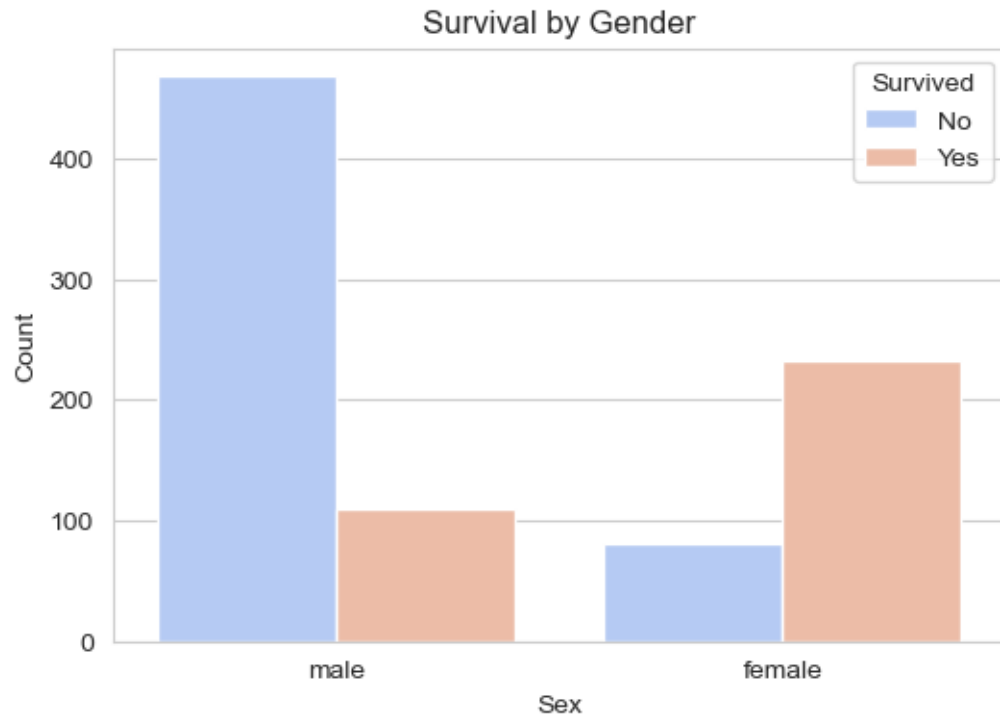
```
[6]: plt.figure(figsize=(6,4))
sns.countplot(data=titanic, x="survived", hue="survived", palette="coolwarm",
             legend=False)
plt.title("Survival Count")
plt.xlabel("Survived (0 = No, 1 = Yes)")
plt.ylabel("Count")
plt.show()
```



```
[7]: # This tells you how many people survived vs. died.
      # If more died (which is expected), the "0" bar will be taller or vice-versa.
      # It gives a quick overview of the tragedy's scale.
```

1.5 2. Survival by Gender

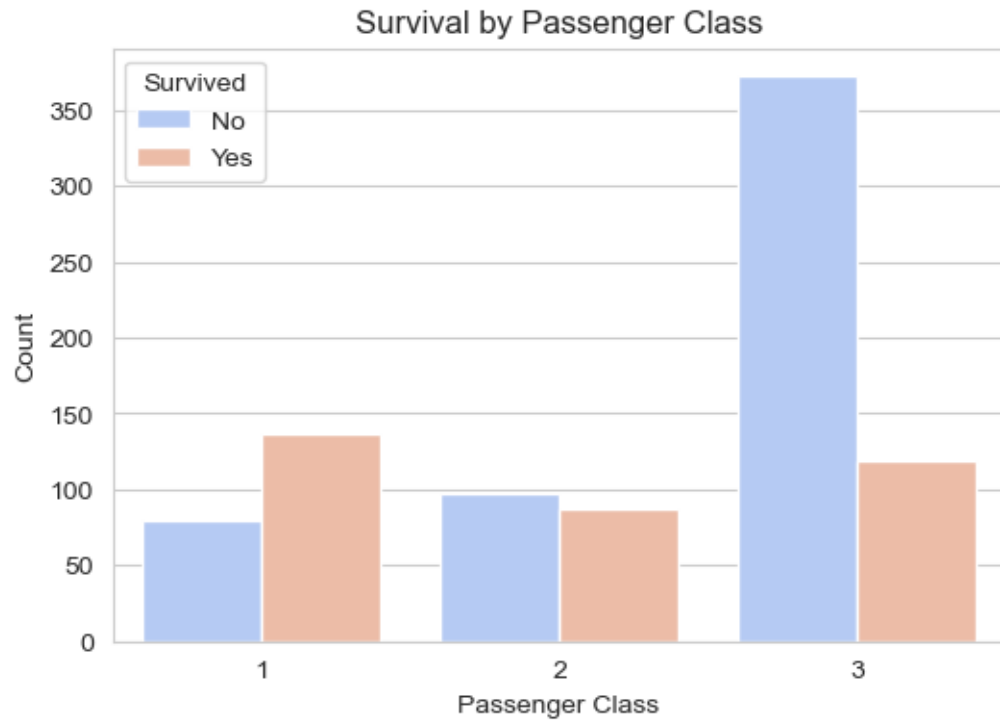
```
[8]: plt.figure(figsize=(6,4))
sns.countplot(x="sex", hue="survived", data=titanic, palette="coolwarm")
plt.title("Survival by Gender")
plt.xlabel("Sex")
plt.ylabel("Count")
plt.legend(title="Survived", labels=["No", "Yes"])
plt.show()
```



```
[9]: # Look at survival difference between males and females.
      # In Titanic, females had higher survival rates due to "women and children
      ↪first."
      # If one gender has a much higher "Yes" bar, that's a survival bias.
```

1.6 3. Survival by Passenger Class

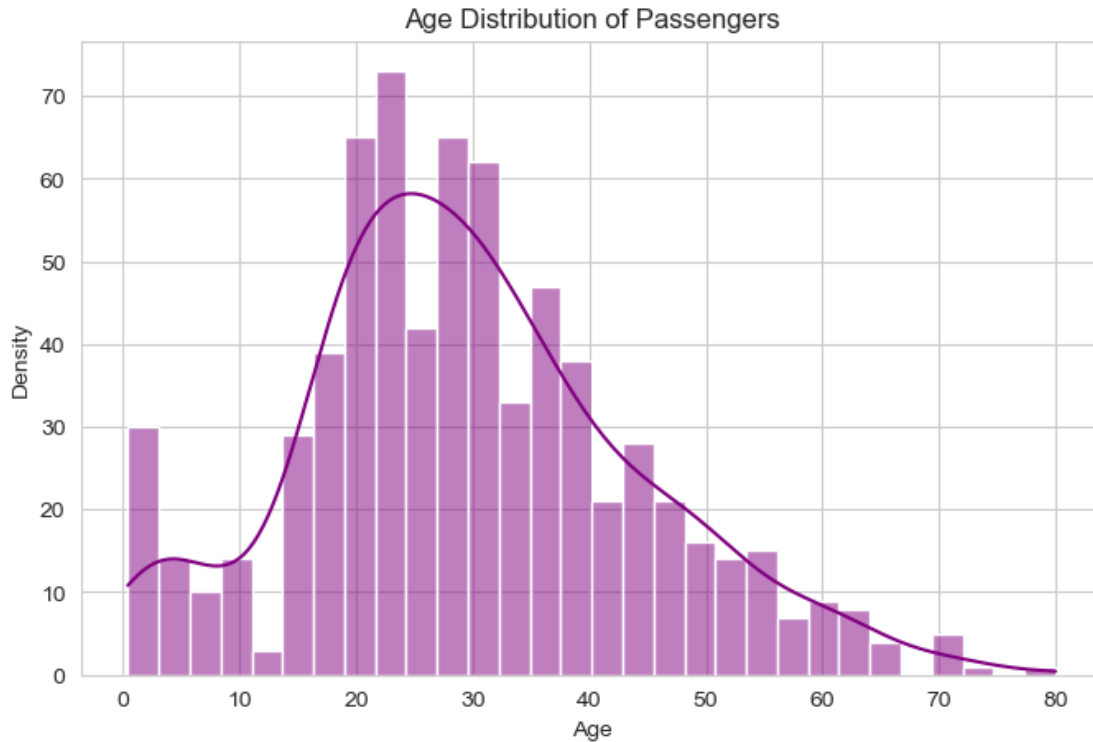
```
[10]: plt.figure(figsize=(6,4))
      sns.countplot(x="pclass", hue="survived", data=titanic, palette="coolwarm")
      plt.title("Survival by Passenger Class")
      plt.xlabel("Passenger Class")
      plt.ylabel("Count")
      plt.legend(title="Survived", labels=["No", "Yes"])
      plt.show()
```



```
[11]: # See if wealth/class affected survival.
      # 1st class likely has more survivors.
      # If 3rd class has more deaths, this may reflect social inequality at the time.
```

1.7 4. Age Distribution of Passengers

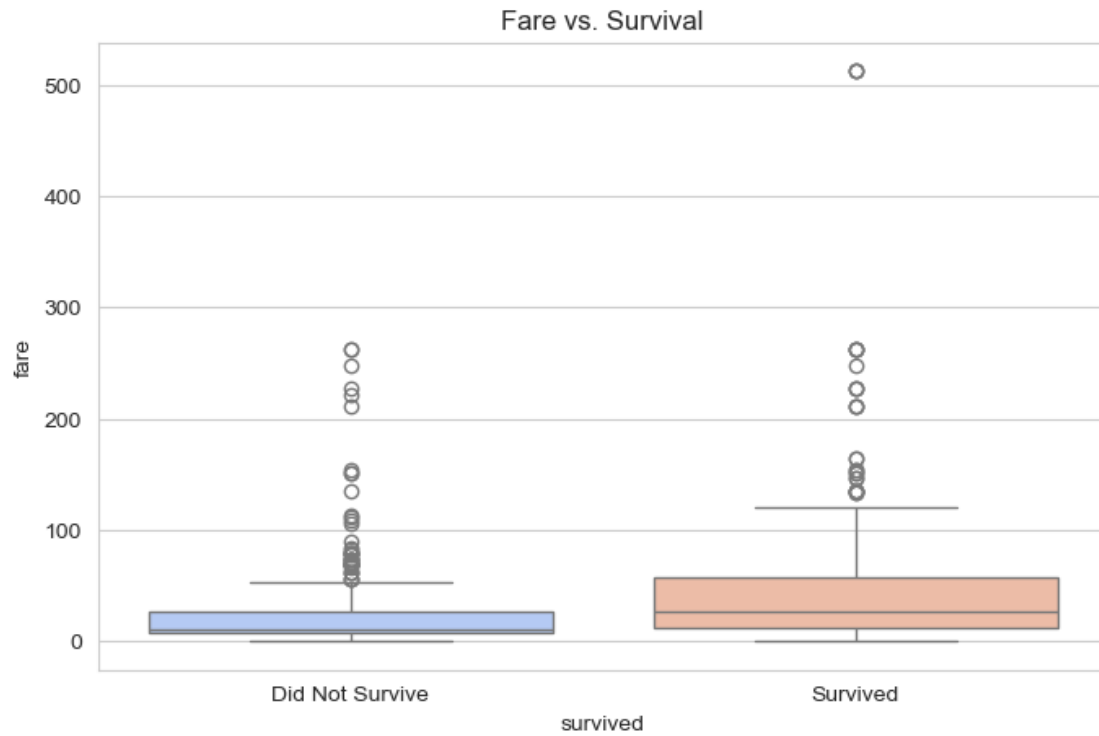
```
[12]: plt.figure(figsize=(8,5))
      sns.histplot(titanic["age"].dropna(), kde=True, bins=30, color="purple")
      plt.title("Age Distribution of Passengers")
      plt.xlabel("Age")
      plt.ylabel("Density")
      plt.show()
```



```
[13]: # Spot age groups with higher survival.
      # Look for younger passengers, especially children, surviving more.
      # Also helps you decide if you want to create age groups (like "child",
      ↪ "adult", etc.)
```

1.8 5. Survival based on Fare Paid

```
[14]: plt.figure(figsize=(8,5))
      sns.boxplot(data=titanic, x="survived", y="fare", hue="survived",
      ↪ palette="coolwarm", legend=False)
      plt.title("Fare vs. Survival")
      plt.xticks([0, 1], ["Did Not Survive", "Survived"])
      plt.show()
```



```
[15]: # Boxplots show fare distribution - min, max, median, and outliers.
      # If survivors paid more on average, you'll see higher fare values for survived,
      ↳ = 1.
      # Tells you if money increased chances of survival (linked to class too).
```